PROJECT TRASURE • OFFICIAL SYSTEM DOCUMENTATION

USER MANUAL

Project TRASURE: B2B Circular Economy Auction Engine & By-Product Trade Platform (Web Platform v1.0)

Standard Operating Procedure & End-User Reference Guide

Course: Software Engineering Laboratory & Theory

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1. System Overview & Architectural Concept

Project TRASURE is an enterprise B2B circular economy platform developed to streamline the sourcing, freight logistics estimation, dynamic reverse auctioning, and statutory environmental compliance of industrial bulk by-products. Heavy process industries—such as thermal power stations, chemical manufacturing plants, fertilizer complexes, and metallurgical refineries—generate massive volumes of secondary materials (including Fly Ash, Ground Granulated Blast-Furnace Slag [GGBS], Spent Sulfuric Acid, and Phosphogypsum).

TRASURE provides industrial procurement managers and plant sustainability heads with a centralized, transparent web console to:

1. **Discover True Delivered Landed Economics:** Avoid misleading ex-factory rates by dynamically calculating delivered landed prices that factor actual highway transit distance, heavy freight tariffs (₹/km/MT), and terminal handling fees.
2. **Visualize Real Road Transit Corridors:** Inspect turn-by-turn GIS highway routes plotted directly between source factories and destination hubs with real-time transit hour estimates.
3. **Execute Live B2B Reverse Auctions:** Procure bulk by-products at competitive market prices through a dynamic reverse auction engine protected by automated anti-sniping extensions.
4. **Automate SPCB Regulatory Compliance:** Seamlessly verify State Pollution Control Board (SPCB) consent credentials and automatically generate standardized Form 10 Hazardous Waste Movement Manifests.
5. **Generate Commercial B2B Tax Invoices:** Automatically compute statutory taxes (18% GST split into 9% CGST and 9% SGST) and produce verifiable commercial invoices.

Client-Side Single Page Application Architecture

Project TRASURE is constructed entirely as a lightweight, zero-dependency **Single Page Application (SPA)** using standards-compliant HTML5, responsive CSS3, and modern Vanilla ES6+ JavaScript. It executes directly inside any modern web browser without requiring complex backend runtime servers, external database configurations, or heavy framework installations.

2. System Requirements & Getting Started

Hardware & Software Prerequisites

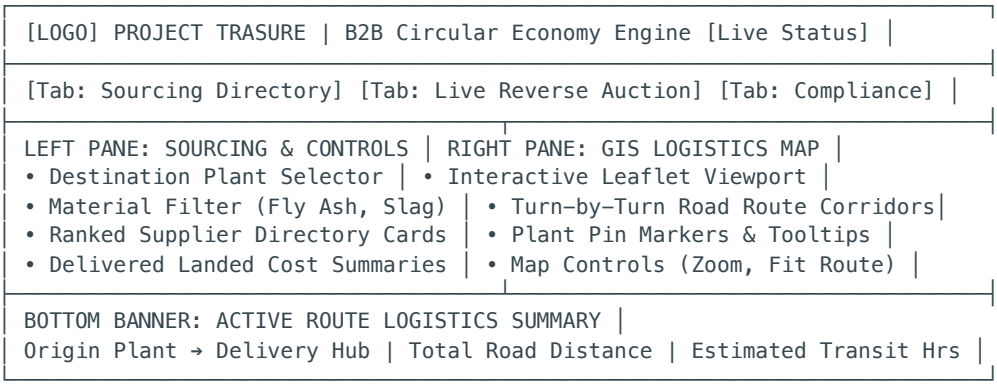
Component	Specification / Requirement
Operating System	Windows 10/11, macOS (10.15+), Linux (Ubuntu, Debian, Fedora), or ChromeOS.
Web Browser	Google Chrome (v90+) [Recommended], Mozilla Firefox (v88+), Apple Safari (v14+), Microsoft Edge (v90+).
Display Resolution	Optimized for desktop displays (1280×720 minimum; 1920×1080 recommended).
Network Connectivity	Minimal internet connection required for external map tile fetching (Esri) and OSRM highway routing. If disconnected, the platform automatically switches to its offline geometric vector routing mode.

How to Launch the Application

1. Extract or locate the Project TRASURE directory on your workstation.
2. Locate the file named `TRASURE_website.html` in the project root directory.
3. Double-click the file to open it in your default web browser, or right-click and choose **Open With > Google Chrome**.
4. The application initializes immediately. On first launch, TRASURE automatically seeds default industrial by-product listings, verified supplier plants, and standard freight parameters into your browser's sandboxed `localStorage`.

3. System Navigation & Operational Layout

The TRASURE web interface is organized into three cohesive visual zones designed for high operational efficiency:



- **Top Navigation Bar:** Displays platform identity, current active receiving hub, and navigation pills to jump between the **Sourcing Directory**, **Live Reverse Auction Exchange**, and **SPCB Compliance Desk**.
- **Left Panel (Operational Desk):** Houses multi-parameter search filters, plant destination selectors, and the ranked by-product supplier catalog cards displaying ex-plant vs. delivered landed rates.
- **Right Panel (GIS Sourcing Map):** Features a dedicated Leaflet-powered GIS viewport displaying the high-contrast Esri World Gray Canvas basemap, supplier plant pins, and active turn-by-turn highway transit corridors.
- **Bottom Summary Ribbon:** Displays an at-a-glance logistics briefing: origin plant name, destination delivery hub, exact road distance in kilometers, estimated truck transit time, and freight cost per kilometer per metric tonne.

4. Core Features & Step-by-Step Procedures

Module 1: Material Sourcing Catalog & Landed Cost Discovery

Enables procurement managers to browse available industrial by-product lots and immediately evaluate them based on true delivered landed cost rather than deceptive ex-factory base prices.

Step-by-Step Sourcing Workflow:

1. Navigate to the **Sourcing Directory** tab.
2. In the **Destination Plant Selector** dropdown, select your receiving facility (e.g., *Vadodara Hub, Gujarat* or *Ramagundam Plant, Telangana*).
3. Use the **Category Filter** pills to isolate specific materials:
    - **All Materials**
    - **Fly Ash** (Class F / Class C thermal power by-product)
    - **GGBS Slag** (Blast furnace cementitious additive)
    - **Spent Acid** (Industrial regenerated sulfuric acid)
    - **Phosphogypsum** (Fertilizer chemical by-product)
4. Inspect the generated supplier cards:
    - **Base Ex-Plant Price:** The raw quotation at the seller's gate.
    - **Calculated Freight:** Transit fee calculated based on exact highway distance.
    - **Delivered Landed Cost:** Highlighted in bold green (e.g., `₹3,239/MT`).
    - The vendor with the lowest delivered cost is automatically badged as **"Optimal Landed Cost"**.
5. Click **"View Route on Map"** on any supplier card to highlight that specific vendor's transit corridor on the GIS map.

Module 2: GIS Logistics Transit Routing & Map Exploration

Provides real-world road network visualization of bulk freight transit, allowing logistics managers to evaluate highway corridors, travel times, and geographic feasibility.

Step-by-Step GIS Exploration:

1. In the GIS Map panel, review the rendered map:
    - **Green Marker:** Your configured destination receiving facility.
    - **Orange Markers:** Verified supplier production plants.
    - **Green Polyline:** The active turn-by-turn road transit corridor.
2. Click **"Fit Route"** to automatically zoom and center the map viewport so both origin and destination facilities are in full view.
3. Click **"My Plant"** to focus directly on your designated delivery terminal.
4. Hover your mouse over any production facility marker to view a high-contrast tooltip displaying plant name, available stock quantity (MT), and landed price.
5. In the bottom **Active Sourcing Route** card, review:
    - **Total Highway Distance:** Measured via real road geometry (e.g., `1395 km`).
    - **Estimated Transit Time:** Standard heavy-haulage truck speed estimation (e.g., `~25 hrs Transit`).
    - **Freight Tariff:** Standard rate per kilometer per metric tonne (e.g., `₹2.01/km/MT`).

Module 3: Live B2B Reverse Auction Exchange & Bidding Desk

Allows bulk industrial buyers to conduct dynamic reverse auctions where pre-qualified industrial sellers compete to fulfill procurement tenders by progressively bidding downwards.

Step-by-Step Bidding Procedure:

1. Click the prominent orange button: **"⚡ Launch & Enter Live B2B Auction Exchange"**.
2. The **Live Reverse Auction Floor Modal** opens:
    - **Tender Reference:** e.g., *Lot #FA-2026-900 (Fly Ash, Grade A - 500 MT)*.
    - **Ceiling Starting Price:** Maximum rate established by the buyer (e.g., `₹4,200/MT`).
    - **Current Leading (Lowest) Bid:** The best market offer submitted so far (e.g., `₹3,800/MT`).
    - **Countdown Clock:** Live ticker timer initialized to `03:00` (180 seconds).
3. **Placing a Competitive Bid:**
    - As an authorized participating supplier, enter your competitive price per metric tonne into the **"Enter Bid Price (₹/MT)"** field.
    - **Rule 1:** The bid must be strictly less than the current leading bid.
    - **Rule 2:** The bid must decrease the price by at least the minimum required decrement (₹50/MT).
    - Click **"Submit Counter-Bid"**.
4. **Anti-Sniping Protection in Action:**
    - If a counter-bid is submitted when the countdown clock has **less than 60 seconds** remaining, the auction engine automatically activates Anti-Sniping Protection.
    - The countdown timer immediately adds **60 seconds** of overtime.
    - A bright amber banner alerts all participants: *"Anti-Sniping Overtime Triggered! +60s Added"*.
5. **Auction Conclusion & Award:**
    - When the countdown timer reaches `00:00`, the auction floor locks automatically.
    - The lowest bidder is declared the winner, and the finalized contract price is locked.
    - Click **"Proceed to Settlement & Compliance"** to generate the commercial invoice and transit documents.

Module 4: SPCB Environmental Compliance & Manifest Desk

Ensures all secondary by-product transactions comply with statutory environmental guidelines established by the Central and State Pollution Control Boards (CPCB/SPCB) under the Hazardous and Other Wastes Rules.

Step-by-Step Manifest Generation:

1. From the main menu, click **"Environmental Compliance"** or click through from a completed auction.
2. Select the transaction lot to audit.
3. Review the compliance parameters verified by the engine:
    - **SPCB Consent to Operate (CTO) ID:** Verified authorization certificate number.
    - **Material Categorization:** Verified whether classified as Non-Hazardous By-Product or Schedule II/III Regulated Waste.
    - **HAZCHEM Schedule Code:** Automated assignment (e.g., *Class 8 Corrosive - UN 1830* for Spent Acid).
4. Enter the authorized logistics transit credentials:
    - **Transporter Agency Name**
    - **Vehicle Registration Number** (e.g., `GJ-06-AX-4821`)
    - **Driver Contact & Emergency Mobile**
5. Click **"Generate SPCB Form 10 Manifest"**.
6. The system compiles a formatted, 6-copy statutory transit manifest ready for printing and physical escort with the transport vehicle.

Module 5: Automated Invoicing & Financial Settlement Desk

Compiles an itemized commercial B2B tax invoice detailing raw material costs, freight charges, and dual GST statutory breakdowns.

Step-by-Step Invoicing Procedure:

1. Upon completing an auction or spot order, click **"View Commercial Tax Invoice"**.
2. Review the structured billing components:
    - **Invoice Metadata:** Unique Tax Invoice Number (e.g., `TRS-2026-904`), issue date, and timestamp.
    - **Consignor (Seller) & Consignee (Buyer):** Legal corporate names, addresses, and GSTIN numbers.
    - **Material Line Item:** Material description, HSN code, ordered quantity in MT, and agreed unit rate.
    - **Logistics Line Item:** Freight haulage charges calculated from highway distance.
    - **Statutory Taxation:** Central GST (CGST @ 9%) + State GST (SGST @ 9%) = Total Statutory GST (18%).
    - **Grand Total Payable:** Displayed clearly in Indian Rupees (₹).
3. Click the **"Print / Download Invoice (PDF)"** button to trigger your browser's native print-to-PDF dialogue.

Module 6: System Administration & Master Data Calibration

Allows administrators or evaluators to recalibrate simulation parameters, reset test transactions, and switch active receiving facilities.

Administrative Maintenance Steps:

1. Click the **"Settings / Master Data"** link in the platform footer.
2. To simulate different geographic logistics scenarios:
    - Choose a different default buyer terminal (e.g., *Vadodara, Chennai, Panipat, or Ramagundam*).
    - Adjust the **Base Freight Tariff Slider** (default: `₹2.00 / km / MT`).
3. To reset the platform to its clean initial state:
    - Click the **"Reset to Factory Seed Data"** button.
    - Confirm the prompt; TRASURE will clear browser `localStorage` and remount clean mock listings, reset auction clocks, and re-index supplier coordinates.

5. Troubleshooting & Frequently Asked Questions (FAQ)

Frequently Asked Questions

Q1: Can Project TRASURE run offline without an active internet connection?
Answer: **Yes.** TRASURE is architected as an offline-resilient Single Page Application. All core modules—including catalog filtering, landed cost calculations, reverse auction bidding, anti-sniping timers, SPCB manifest generation, and invoice billing—operate 100% locally. If the browser cannot reach external map tile servers, the GIS engine automatically plots a fallback vector path between origin and destination coordinates without interrupting workflow.

Q2: Why does the Delivered Landed Cost differ significantly from the Ex-Plant price?
Answer: Secondary industrial materials are heavy bulk goods with low value-to-weight ratios. Sourcing Fly Ash from Jharsuguda (Odisha) to Vadodara (Gujarat) entails over 1,300 km of road transport, adding approximately `₹2,700/MT` in freight. TRASURE deliberately factors this in to protect buyers from unexpected landed expenses.

Q3: How does the Anti-Sniping rule protect participating bidders?
Answer: In unmanaged auctions, predatory bots submit bids in the final millisecond, giving human bidders no time to counter. TRASURE enforces an automated 60-second overtime rule: any valid bid received with less than 60 seconds remaining resets the clock by adding 60 seconds, ensuring fair, transparent price discovery.

Common Troubleshooting Scenarios

Issue Observed	Probable Cause	Recommended Resolution
Map shows gray grid or fails to render road path	Browser blocked external tile fetch or network is throttled.	Click the **"Fit Route"** button. If offline, the platform automatically renders the geometric fallback corridor.
Counter-bid rejected on submission	Entered bid does not meet the minimum decrement of ₹50/MT below the current leader.	Ensure your bid is at least ₹50 lower than the current displayed lowest bid.
Invoice displays NaN or missing totals	LocalStorage state was interrupted or corrupted during a browser reload.	Click **"Reset to Factory Seed Data"** in the footer to restore pristine seed values.
Map marker labels overlapping on small screens	Zoom level too low for closely located industrial facilities.	Click the `+` zoom button on the top-left of the map or click **"Fit Route"** to space out markers.

6. Technical Support & Feedback

For technical support, software verification questions, or academic inquiries regarding Project TRASURE:

Contact Item	Details
Lead Architect & Developer	**Niv Sorathiya**
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GitHub Repository	https://github.com/nivsorathiya-web/Project-TRASURE-B2B-Circular-Economy-Engine